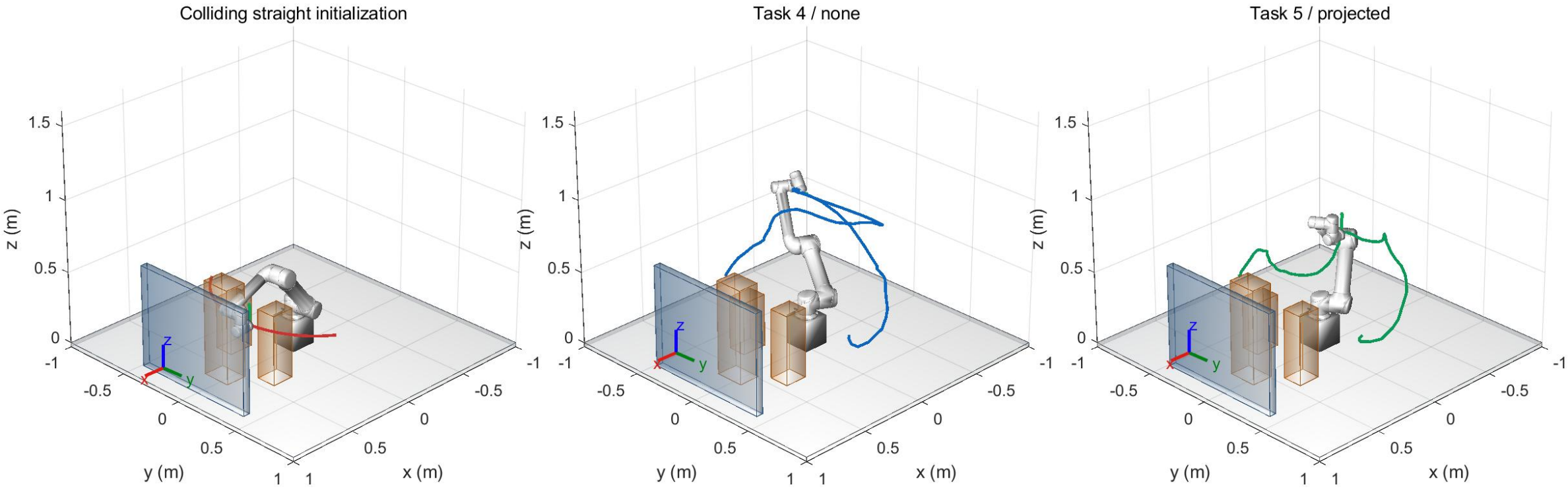
UR5 transfer above a table



A fixed pedestal, table and five shared obstacle volumes

New upright endpoints; the common straight initialization collides at 92/181 states

Upright orientation and collision acceptance

Task 4: self-aware STOMP
Task 5: same planner + orientation penalty and PoE projection

Tool y follows world +z; yaw remains free
Orientation weight 1000; maximum accepted tilt 5 degrees

Both: exact endpoints, limits, 35 mm proxy gap,
zero unpermitted world collisions and zero self collisions

Only the fixed pedestal may contact the table. Every UR5 link and tool is checked.

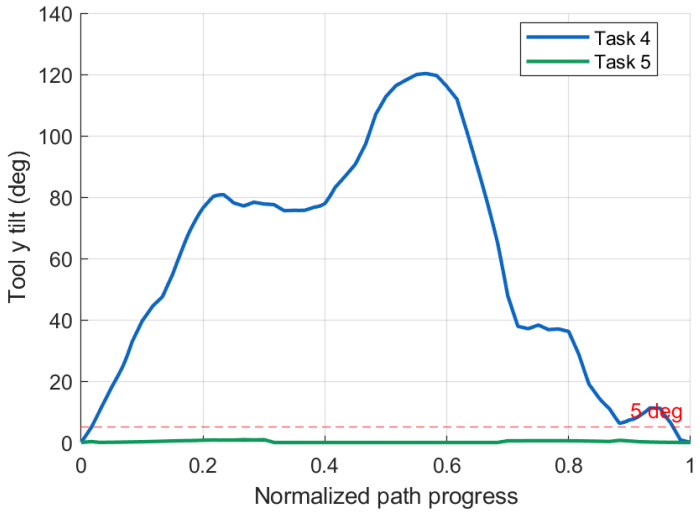
Reserved-seed workcell results

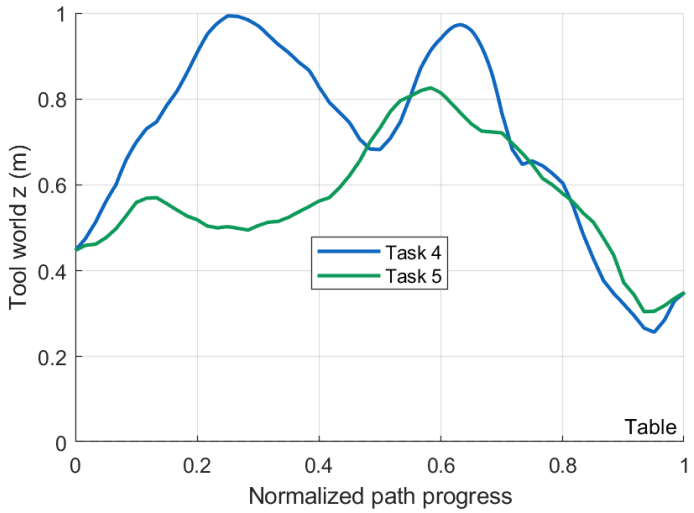
Condition	Success	Median time (s)	Worst tilt (deg)	Min gap (mm)	Min tool z (mm)
Task 4	20/20	30.42	169.591	35.41	273.1
Task 5	20/20	68.19	0.997	35.23	280.9

20 held-out seeds, 40 runs, 20 common successful pairs
Extrema use the additional 0.00625 rad check

Fixed scene and finite seeds; this is not a general reliability guarantee

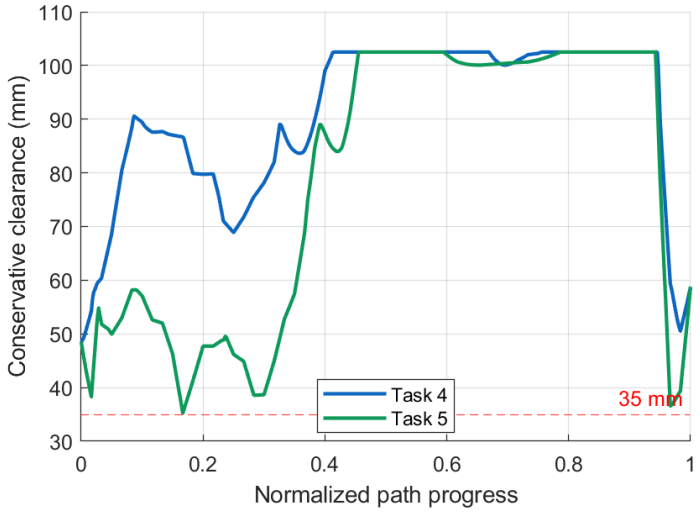
Paired demonstration and study limits

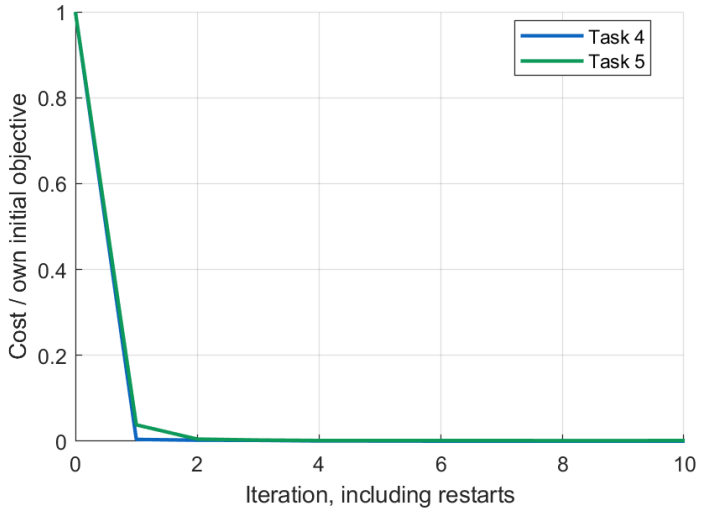




Task 5 worst formal tilt
0.997 degrees

All formal tools stay
above the table





Discrete checks
No payload or dynamics
Progress is not execution time